

Proof attempt — infinitely many primes

Mode: CERTIFICATION candidate (this artifact is a *proof attempt*; it carries no proof weight until a fresh `math-nl-verifier` accepts it). The author does **not** certify this result.

Problem: `infinitely-many-primes`. **Target (verbatim from `request.md`):**

| Prove that there are infinitely many primes.

Inputs used: `request.md`, `nl/sketcher/target-contract.md`, `nl/sketcher/lemma-plan.md`, `nl/sketcher/obligations.md`. Definitions, domain, and normalizations below are taken from the contract; the lemmas `L1` – `L6` are the plan's statements but every one of them is proved here from scratch. Nothing is cited as established merely because the plan asserts it.

0. Target, domain, definitions, framework

0.1 THE ASSERTION AND ITS TWO READINGS

Let

$$P := \{ p \in \mathbb{N} : p \text{ is prime} \}.$$

Primary reading (cardinality). `P` is infinite, i.e. `P` is not finite (definition of finite in §0.4, Definition F).

Working form (unboundedness). $\forall n \in \mathbb{N}, \exists p \in P \text{ with } p > n$.

Per `target-contract.md` §1 these two readings are equivalent for subsets of $\mathbb{N}$; that equivalence is *proved* here (Lemma L5 and Lemma L5c), not assumed. The quantifier order of the working form is `$\forall n \exists p$` ; the swapped form `$\exists p \forall n, p > n$` is **false** and is

not used anywhere. The argument below proves the working form and then converts it to the primary reading through L5.

0.2 DOMAIN (CONTRACT $N1$, $N2$)

$\mathbb{N} = \{1, 2, 3, \dots\}$, the positive integers. Primes are drawn from $\mathbb{N}$.

Remark on $N2$ (whether $0 \in \mathbb{N}$). The argument is written for $\mathbb{N} = \{1, 2, \dots\}$. If instead $0 \in \mathbb{N}$, then: (i) primes are still > 1 by $D2$, so 0 is never prime and the set P is literally unchanged; (ii) the working form at $n = 0$ follows from its value at $n = 1$, since any prime $p > 1$ satisfies $p > 0$ (existence of a prime is Lemma L4); (iii) all divisibility occurrences below have $d \geq 1$ or $n \geq 1$ (checked at each step). Hence both readings of $\mathbb{N}$ yield the same true statement, and the proof for the reading $\mathbb{N} = \{1, 2, \dots\}$ transfers.

0.3 DEFINITIONS (CONTRACT $D1$, $D2$)

$D1$ (divisibility). For $a, b \in \mathbb{N}$: $a \mid b$ iff $\exists c \in \mathbb{N}, b = a \cdot c$.

$D2$ (prime; the normalized definition). $p \in \mathbb{N}$ is **prime** iff

$$p > 1 \quad \text{and} \quad \forall d \in \mathbb{N}, (d \mid p \Rightarrow d = 1 \vee d = p).$$

Consequences used: a prime satisfies $p > 1$, so 1 is not prime; and every prime is ≥ 2 .

0.4 FINITENESS (FOR THE PRIMARY READING)

Definition F. A set S is **finite** iff either $S = \emptyset$, or there exist $k \in \mathbb{N}$ and a bijection $f : \{1, \dots, k\} \rightarrow S$. A set is **infinite** iff it is not finite. This matches `target-contract.md` §1 (“no $k \in \mathbb{N}$ and no bijection $\{1, \dots, k\} \rightarrow P$ ”): since $P \neq \emptyset$ (e.g. $2 \in P$, proved in §1, E11), the empty alternative never applies to P .

0.5 AMBIENT FRAMEWORK FOR $\mathbb{N}$

We use the standard model $\mathbb{N} = \{1, 2, 3, \dots\}$. The following are its standard properties (each a theorem of the Peano axioms / the ordered semiring $\mathbb{N}$; **none mentions primes**, so using them is not circular with respect to the target):

- **(AR1)** $\mathbb{N}$ is a commutative semiring with multiplicative identity 1 ; $1 \leq a$ for every $a \in \mathbb{N}$.
- **(AR2)** $\leq$ is a total order on $\mathbb{N}$ (reflexive, antisymmetric, transitive, and total), compatible with arithmetic: $a \leq b \Rightarrow a+c \leq b+c$ and $ac \leq bc$.
- **(AR3)** $a < b \Leftrightarrow \exists c \in \mathbb{N}, b = a + c$; and (discreteness) $a < b \Rightarrow a+1 \leq b$.
- **(AR4)** Addition and multiplication cancel: $a+c = b+c \Rightarrow a = b$, and $ac = bc \Rightarrow a = b$.
- **(AR5)** Well-ordering: every nonempty $S \subseteq \mathbb{N}$ has a least element.

All invocations of these are flagged as 0-T1 (AR5) and 0-T2 / 0-AR* in the ledger. No other infinity/factorization principle is used anywhere.

1. Elementary facts on $\mathbb{N}$, order, divisibility, and factorial

Every fact is proved from §0.5. Notation: $a > b$ means $b < a$; $a \geq b$ means $b \leq a$.

E1 (trichotomy / totality). For $a, b \in \mathbb{N}$, exactly one of $a < b$, $a = b$, $a > b$ holds; in particular $a \leq b$ or $b \leq a$. *Proof.* $\leq$ is a total order (AR2), so $a \leq b$ or $b \leq a$. Define $<$ by $a < b \Leftrightarrow a \leq b \wedge a \neq b$ (AR3 gives the equivalent $\exists c, b = a+c$). If $a \leq b$ and $b \leq a$, antisymmetry gives $a = b$; otherwise exactly one of $a < b$, $b < a$ holds. ■

E2 (discreteness, restated). $a < b \Rightarrow a+1 \leq b$ (AR3).

E3 (product dominates factors). For $a, b \in \mathbb{N}$: $ab \geq a$ and $ab \geq b$. *Proof.* $b \geq 1$ (AR1). From $b \geq 1$, multiply by a : $ab \geq a \cdot 1 = a$ (AR2 + AR1). Symmetrically with $a \geq 1$. ■

E4 (no factor 1 unless both are 1). If $a, b \in \mathbb{N}$ and $ab = 1$, then $a = b = 1$. *Proof.* Suppose $a \neq 1$. Since $a \geq 1$ (AR1) and $a \neq 1$, E2 gives $a \geq 2$. Then $ab \geq a \geq 2 > 1$ (E3), contradicting $ab = 1$. Hence $a = 1$; symmetrically $b = 1$. ■

E5 (cancellation). $ac = bc \Rightarrow a = b$ (AR4). Also used: $a+c = b+c \Rightarrow a=b$ (AR4).

E6 (divisibility basics). For $a, b, c \in \mathbb{N}$: (a) $a \mid a$; (b) $1 \mid b$; (c) (transitivity) $a \mid b$ and $b \mid c \Rightarrow a \mid c$. *Proof.* (a) $a = a \cdot 1$. (b) $b = 1 \cdot b$. (c) $b = a \cdot x$, $c = b \cdot y$ with $x, y \in \mathbb{N}$; then $c = a(xy)$ and $xy \in \mathbb{N}$. ■

E7 ($\mid$ implies $\leq$). If $a \mid b$ then $a \leq b$. Consequently, if $a \mid b$ and $b \mid a$ then $a = b$. *Proof.* $b = a \cdot c$ with $c \in \mathbb{N}$, $c \geq 1$; so $b = ac \geq a \cdot 1 = a$ by E3/AR2. For the consequence apply E7 twice and antisymmetry. ■

E8 (only 1 divides 1). If $d \in \mathbb{N}$ and $d \mid 1$ then $d = 1$. *Proof.* $1 = d \cdot c$ with $c \in \mathbb{N}$; E4 gives $d = 1$. ■

E9 (divisibility is linear under subtraction). Let $a, b, d \in \mathbb{N}$ with $d \mid a$, $d \mid b$, and $a > b$. Then $d \mid (a - b)$, where $a - b$ is the unique $c \in \mathbb{N}$ with $a = b + c$ (it exists because $a > b$, and is unique by E5). *Proof.* Write $a = d \cdot x$, $b = d \cdot y$ with $x, y \in \mathbb{N}$. If $x \leq y$ then $a = dx \leq dy = b$ (AR2), contradicting $a > b$. Hence by E1 $x > y$, so $x = y + c'$ for some $c' \in \mathbb{N}$. Then $a = dx = d(y + c') = dy + dc' = b + dc'$. By uniqueness of the difference, $a - b = dc'$, so $d \mid (a - b)$. ■

E10 (factorial). Define $n!$ for $n \in \mathbb{N}$ by recursion: $1! := 1$, and $(n+1)! := n! \cdot (n+1)$ for $n \geq 1$. Then for all $n \in \mathbb{N}$: (a) $n! \geq 1$; (b) $n! + 1 \geq 2 > 1$. *Proof.* (a) Induction: $1! = 1 \geq 1$; and $(n+1)! = n! \cdot (n+1) \geq n! \geq 1$ by E3 and the induction hypothesis. (b) From (a), $n! + 1 \geq 1 + 1 = 2 > 1$. ■

E11 (2 is prime). $2 \in P$; hence $P \neq \emptyset$. *Proof.* $2 > 1$. Let $d \in \mathbb{N}$ with $d \mid 2$. By E7, $d \leq 2$. Since $d \geq 1$ and $1 \leq d \leq 2$, E2 (applied to $1 < d$ when $d > 1$) gives $d = 1$ or $d = 2$. Hence every divisor of 2 in $\mathbb{N}$ is 1 or 2, so 2 is prime by D2. ■

2. Lemma L1 — Every integer > 1 has a prime divisor

Statement (L1). For every $m \in \mathbb{N}$ with $m \geq 2$ there exists $p \in \mathbb{N}$ such that $p \mid m$ and p is prime (D2).

Proof. Fix $m \in \mathbb{N}$ with $m \geq 2$. Define

$$S := \{ d \in \mathbb{N} : d \mid m \text{ and } d > 1 \}.$$

Nonemptiness. $m \mid m$ by E6(a), and $m > 1$ since $m \geq 2$. Hence $m \in S$, so $S \neq \emptyset$. (Precondition of AR5 met: S is a nonempty subset of $\mathbb{N}$.)

Choice of minimal element. Apply **AR5 (well-ordering of $\mathbb{N}$; obligation 0–T1)**: S has a least element. Set

$$p := \min S,$$

so $p \in S$; in particular $p \mid m$ and $p > 1$.

p is prime. $p > 1$ holds. Let $d \in \mathbb{N}$ with $d \mid p$; we must show $d = 1 \vee d = p$. By E7, $d \leq p$. By E1, either $d \leq 1$ — and then $d = 1$ since $d \geq 1$ — or $d > 1$. In the second case $d \mid p$ and $p \mid m$ give $d \mid m$ by **E6(c) (transitivity of $\mid$)**; together with $d > 1$ this says $d \in S$. As p is the least element of S , $p \leq d$. With $d \leq p$ and antisymmetry, $d = p$. Either way $d = 1$ or $d = p$, so the D2 divisor condition holds and p is prime. ■

Non-circularity check (0–L1, keystone). The proof uses **only**: the definition D1 of $\mid$; reflexivity (E6a) and transitivity (E6c) of $\mid$; $\mid$ -implies- $\leq$ (E7); E1 trichotomy; and **AR5 well-ordering**. It does **not** use the infinitude of the primes (the target), does **not** use the Fundamental Theorem of Arithmetic (existence *or* uniqueness), does **not** use Euclid’s lemma $p \mid ab \Rightarrow p \mid a \vee p \mid b$, and does **not** assume P finite or infinite. The minimal element p is produced directly by well-ordering, not by iterative “extract a prime factor from a proper divisor” (which would presuppose L1). ■

3. Lemma L2 — A prime $\leq n$ divides $n!$

Statement (L2). For all $n \in \mathbb{N}$ and every prime p : if $p \leq n$ then $p \mid n!$.

Proof. Induction on $n \geq 1$ using the recursion of E10.

Base $n = 1$. If p is prime then $p > 1$ (D2), so $p \leq 1$ is impossible (by E1, $p \leq 1$ with $p \geq 1$ forces $p = 1$, contradicting $p > 1$). Hence the statement is vacuously true for $n = 1$.

Inductive step. Assume L2 holds for $n \geq 1$; prove it for $n+1$. Let p be prime with $p \leq n+1$. By **E1 (trichotomy)** either $p \leq n$ or $p > n$.

- If $p \leq n$: the induction hypothesis gives $p \mid n!$. Also $n! \mid (n+1)!$, because the recursion gives $(n+1)! = n! \cdot (n+1)$. By **E6(c) (transitivity of $\mid$)**, $p \mid (n+1)!$.
- If $p > n$: then $n < p \leq n+1$. By **E2 (discreteness)**, $n < p$ gives $n+1 \leq p$; with $p \leq n+1$ and antisymmetry, $p = n+1$. Then $(n+1)! = n! \cdot (n+1) = n! \cdot p$, so $p \mid (n+1)!$.

The two cases are exhaustive (E1) and both yield $p \mid (n+1)!$. ■

Preconditions checked. Factorial is defined for all $n \in \mathbb{N}$ (E10); $p \in \mathbb{N}$ by D2. No hypothesis beyond D1, D2, and the factorial recursion is used.

4. Lemma L3 — $n! + 1$ is coprime to $n!$

Statement (L3). For every $n \in \mathbb{N}$: (i) $n! + 1 > 1$; and (ii) every $d \in \mathbb{N}$ with $d \mid n!$ and $d \mid (n! + 1)$ satisfies $d = 1$.

Proof. Fix $n \in \mathbb{N}$.

(i) By **E10(a)**, $n! \geq 1$, so $n! + 1 \geq 2 > 1$. ■

(ii) Let $d \in \mathbb{N}$ with $d \mid n!$ and $d \mid (n! + 1)$. Put $a := n! + 1$ and $b := n!$. Then $a > b$ (indeed $a = b + 1$), and $b, a, d \in \mathbb{N}$. Apply **E9 (linearity of $\mid$ under subtraction)** with these a, b, d : the preconditions $d \mid a$, $d \mid b$, $a > b$ all hold, so $d \mid (a - b)$. The difference is $a - b = (n! + 1) - n! = 1$ (the unique c with $n! + 1 = n! + c$ is $c = 1$, by E5). Hence $d \mid 1$, and **E8** gives $d = 1$. ■

Note. No gcd or coprimality primitive is introduced; the statement is phrased with $\mid$ alone, as in the plan.

5. Lemma L4 — Unboundedness of the primes

Statement (L4). For every $n \in \mathbb{N}$ there exists a prime $p \in \mathbb{N}$ with $p > n$.

Proof. Fix $n \in \mathbb{N}$. Put

$$M := n! + 1.$$

By **E10(b)**, $M \geq 2 > 1$, so the hypothesis $m \geq 2$ of **L1** holds for $m := M$. Apply **L1**: there exists a prime $p \in \mathbb{N}$ with $p \mid M$, i.e. $p \mid (n! + 1)$.

Now apply **E1 (trichotomy)** to the pair (p, n) : either $p \leq n$ or $p > n$. These two cases are exhaustive.

- **Case $p > n$.** Done: p is a prime with $p > n$.
- **Case $p \leq n$.** Then p is prime and $p \leq n$, so **L2** gives $p \mid n!$. Also $p \mid (n! + 1)$. Apply **L3(ii)** with $d := p$ (legitimate since $p \in \mathbb{N}$): $p = 1$. But p is prime, so **D2** gives $p > 1$, contradicting $p = 1$. This case is therefore impossible.

Since the second case is contradictory, the first holds, and there is a prime $p > n$. As $n \in \mathbb{N}$ was arbitrary, $\forall n \in \mathbb{N} \exists p \text{ prime}, p > n$. ■

Hypotheses audit. **L4** is a closed $\forall n \exists p$ statement: no finiteness assumption on P , no bound on p , and no extra hypothesis is introduced. All preconditions of **L1**, **L2**, **L3** are checked above; the only order principle used is **E1** (from **AR2/AR3**).

*Remark (reading **N2**, $0 \in \mathbb{N}$).* If $\mathbb{N}$ were taken to contain 0 , the case $n = 0$ is immediate from the case $n = 1$ (there is a prime $p > 1 > 0$), consistent with **target-contract.md** §3. The recursion-based proof above is stated for $n \geq 1$, which is exactly $\mathbb{N} = \{1, 2, \dots\}$.

6. Lemma L5 — Finite subsets of $\mathbb{N}$ are bounded above; and L5c (converse)

Statement (L5). For every $S \subseteq \mathbb{N}$: if S is finite (Definition F) then S is bounded above, i.e. $\exists N \in \mathbb{N}, \forall s \in S, s \leq N$.

Proof. Two cases.

Case $S = \emptyset$. Take $N := 1 \in \mathbb{N}$. The condition $\forall s \in \emptyset, s \leq 1$ is vacuous. ■

Case $S \neq \emptyset$ *finite*. By Definition F there are $k \in \mathbb{N}$ and a bijection $f : \{1, \dots, k\} \rightarrow S$. We prove the following by induction on $k \geq 1$:

(L5-g) For every $k \in \mathbb{N}$, every $S \subseteq \mathbb{N}$ admitting a bijection $f : \{1, \dots, k\} \rightarrow S$ has a **greatest element**: $\exists g \in S, \forall s \in S, s \leq g$.

Base $k = 1$. $S = \{f(1)\}$; take $g := f(1) \in S \subseteq \mathbb{N}$. For $s \in S$, $s = f(1)$ so $s \leq g$. $\checkmark$

Inductive step ($k \rightarrow k+1$). Assume L5-g for $k \geq 1$. Let $S \subseteq \mathbb{N}$ and $f : \{1, \dots, k+1\} \rightarrow S$ be a bijection. Set $S' := S \setminus \{f(k+1)\}$; the restriction $f|_{\{1, \dots, k\}}$ is a bijection $\{1, \dots, k\} \rightarrow S'$. By the induction hypothesis S' has a greatest element g' . Define $g := \max(g', f(k+1))$, the larger of two elements of $\mathbb{N}$ (so $g \in \mathbb{N}$). For any $s \in S$: either $s = f(k+1)$, giving $s = f(k+1) \leq g$; or $s \in S'$, giving $s \leq g' \leq g$. Hence g is a greatest element of S . $\checkmark$

Applying L5-g to the given S and f yields a greatest element $g \in S \subseteq \mathbb{N}$. Take $N := g \in \mathbb{N}$. For every $s \in S$, $s \leq g = N$. Hence S is bounded above. ■

Contrapositive form (used for the target).

(L5*) For every $S \subseteq \mathbb{N}$: if S is *unbounded above* — i.e. $\forall n \in \mathbb{N} \exists s \in S, s > n$ — then S is infinite (not finite).

Proof. L5* is the literal contrapositive of L5: “finite $\Rightarrow$ bounded above” is logically equivalent to “not bounded above $\Rightarrow$ not finite”; “not bounded above” is the displayed unboundedness condition, and “not finite” is the definition of infinite. ■

Statement (L5c, converse; 0–L5c, non-load-bearing). If $S \subseteq \mathbb{N}$ is bounded above then S is finite.

Proof. Suppose $\exists N \in \mathbb{N}, \forall s \in S, s \leq N$. Since every $s \in S \subseteq \mathbb{N}$ satisfies $s \geq 1$, we get $S \subseteq \{1, \dots, N\}$. We show by induction on $N \geq 1$ that **every** subset of $\{1, \dots, N\}$ is finite.

- Base $N = 1$: a subset of $\{1\}$ is $\emptyset$ or $\{1\}$; $\emptyset$ is finite by Definition F, and $\{1\}$ is finite via the identity bijection $\{1\} \rightarrow \{1\}$.

- Step $N \rightarrow N+1$: let $T \subseteq \{1, \dots, N+1\}$. If $N+1 \notin T$ then $T \subseteq \{1, \dots, N\}$, finite by the induction hypothesis. If $N+1 \in T$, write $T = T' \cup \{N+1\}$ with $T' \subseteq \{1, \dots, N\}$, finite by the induction hypothesis; if $T' = \emptyset$ then $T = \{N+1\}$, finite via $1 \mapsto N+1$; otherwise T' has a bijection $f: \{1, \dots, k\} \rightarrow T'$ (some $k \in \mathbb{N}$) and extending it by $k+1 \mapsto N+1$ is a bijection $\{1, \dots, k+1\} \rightarrow T$, so T is finite. Hence $S \subseteq \{1, \dots, N\}$ is finite. ■

Remark. L5 and L5c together give, for every $S \subseteq \mathbb{N}$: S is infinite $\iff S$ is unbounded above. This is exactly the equivalence asserted in `target-contract.md` §1 / ambiguity A1; it is now proved, so the reading choice is discharged rather than assumed.

7. Lemma L6 — Assembly: the original target

Let $P := \{ p \in \mathbb{N} : p \text{ prime} \}$ (D2).

1. $P \subseteq \mathbb{N}$. Immediate from the definition of P . (0-B1.)
2. **Working form.** By L4, for each $n \in \mathbb{N}$ there is a prime $p \in \mathbb{N}$ with $p > n$. That p satisfies $p \in P$ and $p > n$. Hence $\forall n \in \mathbb{N}, \exists p \in P$ with $p > n$, i.e. P is unbounded above. The $\forall n \exists p$ order of L4 is preserved verbatim; the swapped form $\exists p \forall n$ is **not** produced or used. (0-B3.)
3. **Unbounded $\implies$ infinite.** Apply L5* with $S := P$ (legitimate: $P \subseteq \mathbb{N}$ by step 1, and P is unbounded above by step 2). Conclusion: P is infinite.
4. **Reading bridge.** By `target-contract.md` §1 the primary reading of “there are infinitely many primes” **is** the statement “ P is infinite”. Step 3 therefore establishes the original target. (0-B2.)

Conclusion. $P = \{ p \in \mathbb{N} : p \text{ prime} \}$ is infinite; equivalently, $\forall n \in \mathbb{N} \exists p \in \mathbb{N} (p \text{ prime} \wedge p > n)$. ■

8. Global audit

Hypothesis audit. No hypothesis is added anywhere. The target is an unconditional closed statement. In particular:

- no finiteness assumption on P is made in L4 (the contradiction-style Route B of the plan is deliberately *not* used);
- no genericity, nonzero, or regularity hypothesis is inserted: all uses of divisibility occur with arguments in $\mathbb{N}$, and the “ $d \geq 1$ ”/“ $d > 1$ ” conditions are forced by D2, E1, and E10, as shown;
- the ambient principles (AR1)–(AR5) are standard properties of $\mathbb{N}$ and do not mention primes.

Circularity audit. The dependency graph is $D1, D2, AR^* \rightarrow \{L1, L2, L3, L5(-g)\} \rightarrow L4 \rightarrow L5^* \rightarrow L6$. No lemma uses the target. Specifically:

- L1 uses well-ordering of $\mathbb{N}$, D1, and transitivity of $|$ only (§2);
- L2 uses the factorial recursion and transitivity of $|$ only (§3);
- L3 uses subtraction-linearity of $|$ (E9) and E8 only (§4);
- L4 uses L1, L2, L3 and trichotomy only (§5);
- L5/L5-g use Definition F, induction, and $\max$ on $\mathbb{N}$ only (§6);
- the assembly uses L4, L5*, and the definitional reading bridge only (§7). Explicitly **not** used: infinitude of primes, FTA (existence or uniqueness), Euclid’s lemma, Dirichlet/Bertrand/analytic input.

Quantifier audit. The proven statement is $\forall n \in \mathbb{N} \exists p \in P, p > n$ (unboundedness) and the cardinality statement “ P is infinite”. The false swapped form is never asserted.

Boundary audit. 1 is not prime (D2, $p > 1$); 2 is the least prime (E11); the empty and finite cases of “infinitely many” are handled by L5/L5c and the reduction to the non-finite reading.

9. Status and unresolved issues

Every step above is discharged by an explicit argument from D1, D2, and (AR1)–(AR5); the obligation ledger `obligations.md` maps each named obligation of the plan to its discharge. **No unresolved gap remains in this proof attempt**, apart from the following caveats, which are limitations of scope rather than gaps:

- The proof is written for the reading $\mathbb{N} = \{1, 2, \dots\}$ (N1); the transfer to a reading in which $0 \in \mathbb{N}$ is argued in §0.2/§5 and rests only on $p > 1$ for primes.
- L5c / 0-DEF2 (equivalence $D2 \iff D3$, and $\text{bounded} \iff \text{finite}$) are supplied for completeness / to discharge ambiguity A1; 0-DEF2 is not needed by the route and is not proved here (it is recorded as an optional completeness item in obligations.md).
- Acceptance is **not** claimed by the author. A fresh independent math-nl-verifier must check this candidate in a separate task.